Active sensing explains cortical variability and improves visual information

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What looks like noise may reflect active sensing

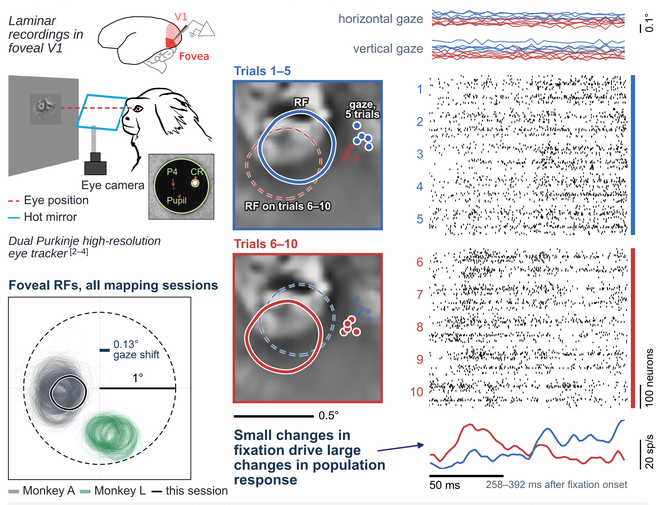
Neurons respond differently across repeated presentations of the same stimulus. These trial-to-trial differences are often described as “noise,” but they can include sensory effects of the animal’s own movements.

Even during fixation, small eye movements continually reshape visual input.

How much cortical variability comes from the eye?

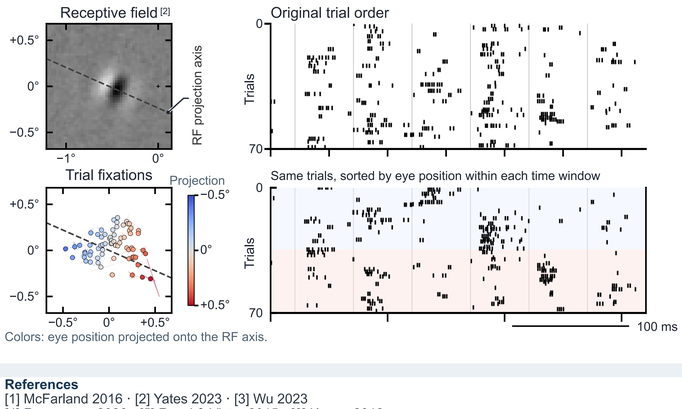
Do fixational eye movements help encode spatial detail?

1 Tiny gaze shifts reshape population responses

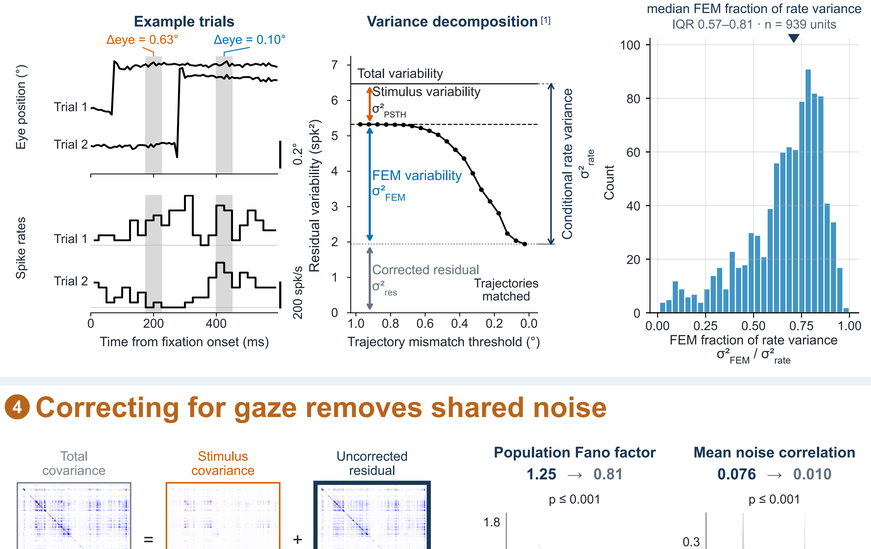


2	1,022	19	20 Hz
awake marmosets	foveal V1 units	recording sessions	repeated natural images

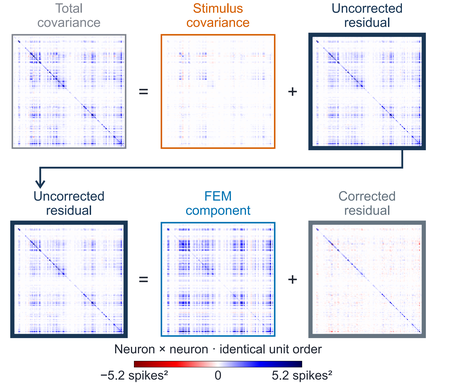
2 One neuron’s “noise” tracks the eye

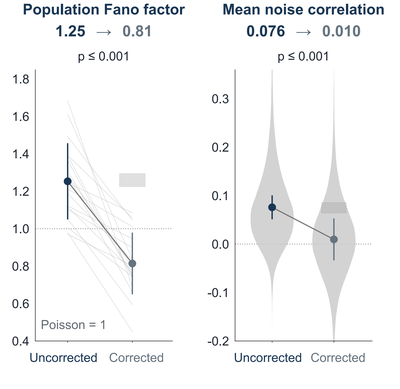


3 Eye movements drive most single-cell rate variance (71%)



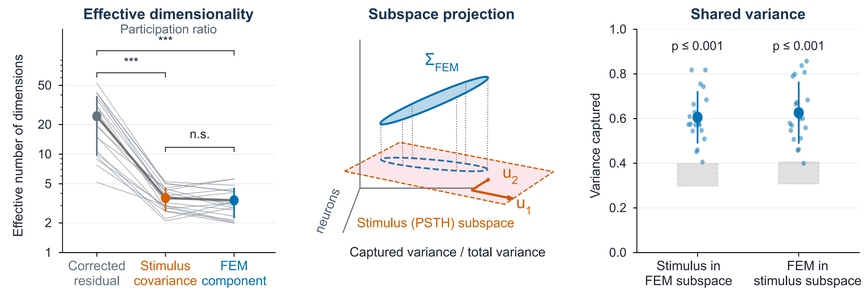
4 Correcting for gaze removes shared noise





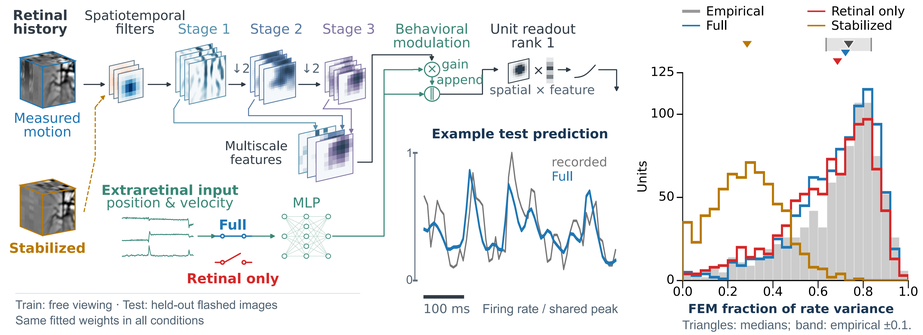
19 sessions; 25-ms bins. Points: mean ± SD. Gray bands: 95% shuffle null; violins: pairwise correlations.
Corrected mean noise correlation: 95% CI −0.010 to 0.029. Both tests compare correction with shuffle.

5 Eye-driven variability follows stimulus dimensions



*** p < 0.001; n.s. p ≥ 0.05
25-ms counts; alignment uses four leading dimensions per component.
Dots: sessions · large dots: mean ± SD
Gray bands: 95% shuffle-null intervals

6 Gaze-dependent retinal input predicts single trials



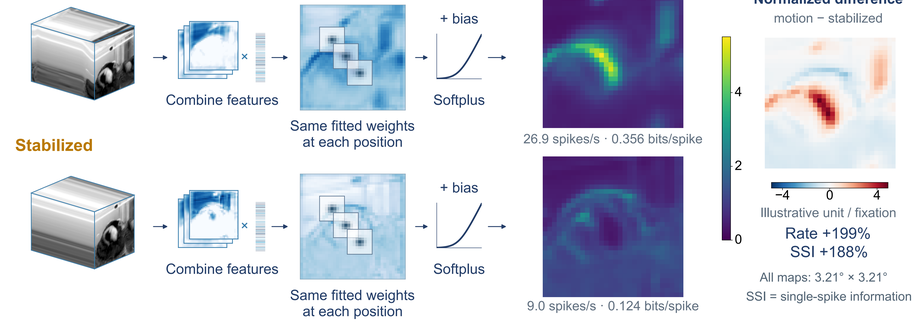
Train: free viewing · Test: held-out flashed images
Same fitted weights in all conditions

Triangles: medians; band: empirical ±0.1.

Median fraction of conditional rate variance explained
962 units · 19 sessions

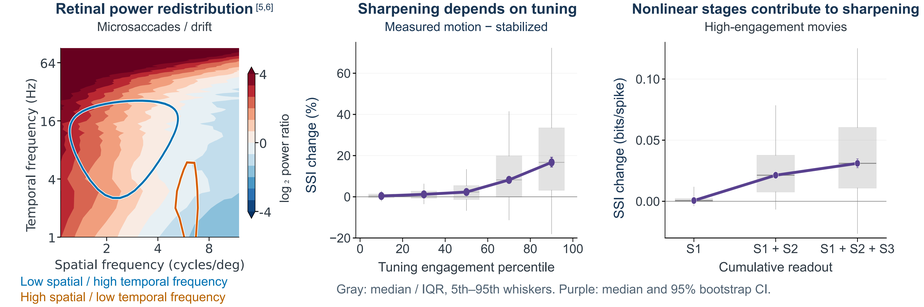
Trial average	Full	Retinal only	Stabilized
0.139	0.235	0.226	0.042

7 In the model, eye motion sharpens spatial coding



8 Sharpening grows with tuning and model depth

725 units



Gray: median / IQR, 5th–95th whiskers. Purple: median and 95% bootstrap CI.
Tuning engagement: within-unit percentile of motion-induced tuning-relevant power.
All quintiles/stages > 0 (Holm-adjusted p < 0.05).

References
[1] McFarland 2016 · [2] Yates 2023 · [3] Wu 2023
[4] Ressmeyer 2026 · [5] Rucci & Victor 2015 · [6] Kuang 2012

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